SAGNIK CHAKRABORTY

BS-MS Student
IISER Kolkata

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SKILLS

- Python
- C++ (Arduino)
- MPI (parallel computing)
- NumPy / SciPy / Matplotlib
- Data analysis & curve fitting
- Origin
- LaTeX
- Arduino / embedded

INTERESTS

- Experimental Physics
- Nuclear Physics
- Computational Physics
- Quantum Mechanics
- Electrodynamics

LANGUAGES

- English
- Hindi
- Bengali

IISER KOLKATA

EXPLORING PHYSICS,
COMPUTATION AND CURIOSITY

PHYSICS
COMPUTATION
EXPERIMENTATION
CURIOSITY

SUMMARY

Second-year BS-MS physics student at IISER Kolkata, drawn to experimental and computational physics. Comfortable with Python-based numerical simulation and data analysis, and building hands-on experience through lab projects in nuclear physics and electronics. Eager to learn and contribute on a research internship in experimental or computational physics.

EDUCATION

Indian Institute of Science Education and Research (IISER) Kolkata
BS-MS Dual Degree Programme
Currently in 2nd Year (3rd Semester)

WORKSHOPS & RESEARCH EXPERIENCE

Gamma–Gamma Coincidence (Micro Project)
Under Prof. Satyabrata Raj, IISER Kolkata

- Performed γ–γ coincidence experiments with ^{60}Co and ^{137}Cs sources using NaI(Tl) (2″×2″, 3″×3″), CdZnTe, and HPGe detectors.
- Acquired and analysed energy spectra via a multichannel analyzer (MCA); calibrated peaks, measured FWHM, and computed energy resolution for each detector.
- Characterised and subtracted Compton background to isolate coincidence peaks, using Python and Origin for fitting and analysis.
- Compared detector performance (resolution vs. efficiency) across NaI(Tl), CdZnTe, and HPGe; tuned high-voltage and amplifier settings.

Parallelization with MPI in Python (Workshop)
Under Prof. Sudip Kr Garain
Singularity: The Astronomy Club, IISER Kolkata

- Implemented parallel numerical routines in Python using mpi4py, distributing computation across multiple processes.
- Completed all assigned exercises and benchmarked parallel vs. serial runtimes to study speedup.

SPARC Workshop on Nanobiophotonics
IISER Kolkata, 12 October 2026

Online Introductory School on Astronomy and Astrophysics
IISER Mohali (Online)

- Completed an online introductory school covering foundational topics in astronomy and astrophysics.

PROJECTS

Time Evolution of Waves in Classical and Quantum Physics

- Built Python simulations of classical waves (strings, standing waves, wave packets) and quantum wave-function evolution via numerical solution of the Schrödinger equation.
- Implemented time-stepping solvers and probability-density visualizations with NumPy and Matplotlib.

Robotics and Embedded Systems

- Built and programmed autonomous robot-car prototypes using Arduino and C++ with motor drivers.
- Implemented wireless communication with nRF24L01 modules and explored Wi-Fi access-point and range-extender modes.